FOOD COMPASS SCORE-10: VALIDATION OF A METHOD FOR EVALUATING THE HEALTHFULNESS OF FOODS AND BEVERAGES USING INGREDIENT LIST INFORMATION

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Abbreviations: FCS, Food Compass Score; FCS-10, Food Compass Score-10; FNDDS; Food and Nutrition Database for Dietary Studies; FPS, food profiling system; GBFDB, Global Branded Food Products Database; GTIN, Global Trade Item Number; NHANES, National Health and Nutrition Examination Survey; NPS, nutrient profiling system; NPV, negative predictive value; PPV, positive predictive value; RMSE, root mean squared error; USDA, United States Department of Agriculture; WWEIA, What We Eat In America.

ABSTRACT

Background: The Food Compass, a novel food profiling system, provides a holistic, validated assessment of healthfulness of foods, beverages and meals using 54 attributes across 9 domains. However, information on several of these attributes is not commonly available.

Objective: We aimed to develop and validate an approach, Food Compass Score-10 (FCS-10), to estimate Food Compass scores using information commonly available on package labels.

Methods: Missing attributes were calculated using weighted scores of each product's ingredients, derived from a dataset of ~10,000 foods and beverages. The final FCS-10 was scaled from 1 (least healthful) to 10 (most healthful). As part of this validation study, diagnostic accuracy analysis was conducted to evaluate the performance of the FCS-10 compared to the original score. Sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were calculated by comparing the FCS-10 recommendation categorisations with the FCS recommendation categorisations (≥7 for foods to encourage, 4-6 for foods to consume in moderation, ≤3 for foods to limit).

Results: FCS-10 produced scores within 1 unit of the original score (when rescaled 1-10 for comparison) for 89% of products (n=481); none deviated >2 units. The correlation between FCS-10 and the original score was high (*r*=0.93). FCS-10 also performed well in identifying products to encourage, moderate, or limit, with overall sensitivity and specificity of 87% and 93%, respectively.

Conclusions: FCS-10 offers a practical approach for estimating the healthfulness of diverse packaged foods and beverages using readily available label data while maintaining the strengths of the original system.

Key words: Food Compass, nutrient profiling, food classification, nutrition, validation study

INTRODUCTION

Unhealthy dietary patterns are a leading cause of disease, disability, and death globally.(1) More effective individual and population-based nutrition strategies are necessary to facilitate healthier dietary behaviors and mitigate the harms of diet-related illnesses. Increasingly, a common approach to support such strategies are nutrient profiling systems (NPSs), algorithms that quantitatively assess and rank the overall healthfulness of foods and beverages. NPSs are being widely applied by governments, food manufacturers, supermarket retailers, and investor coalitions in mandatory national and voluntary industry front-of-pack and menu labelling, establishment of thresholds to regulate food marketing and use of health claims, formulation of food procurement policies in schools and institutions, setting targets for reformulating product portfolios, and making decisions around capital investments and divestments.(2, 3)

Most NPSs have focused on the amount and presence of a few nutrients, such as calories, fat, saturated fat, salt, and sugar, and a few food groups, like fruits and vegetables.(4) However, because this approach omits other critical characteristics of healthfulness (or unhealthfulness), contemporary NPSs are needed to incorporate additional important characteristics such as amounts of protective and harmful food ingredients, extent of processing, and the presence of bioactives and additives.(5) For such systems, a more accurate term may be *food* profiling systems (FPSs). The Food Compass FPS encompasses 54 attributes across nine domains to provide a more holistic assessment of foods and beverages as well as mixed foods, menu items, and meals, while preventing any single attribute or domain from having excessive influence (for example, limiting high scores due primarily to high amounts of nutrient fortification).(5) Food Compass also scores items per 100 kcal, rather than by weight as in common NPSs, which reduces confounding and contradictory scoring from water content (for example, preventing sugar-sweetened beverages from receiving high scores).(6) Food Compass has been recently updated to incorporate more nuanced and valid scoring for food processing, food ingredients, and other factors.(7) The validity of Food Compass has been demonstrated for scoring foods, beverages, and mixed meals, as compared to common NPSs,(5) and as applied to the diets of individuals in relation to health risk factors, prevalent diseases, and prospective risk of total mortality, cardiovascular disease, and obesity.(8-11) In addition, findings from a recent randomized, controlled real-choice experiment revealed that the presence of a Food Compass front-of-pack score reduced the selection of unhealthy foods, increased the selection of healthy foods, and increased overall purchasing (personal communication, Dr. Sean Cash, Tufts University, July 1, 2024).

Yet, while Food Compass is a highly promising FPS that could have widespread and impactful application, several attributes are not commonly available on mandatory nutrition facts panels or ingredients lists. This data gap limits its broad application to the full range of manufactured food products globally. New approaches are needed to enable estimation of the Food Compass score with more commonly available information to optimise its practical utility, without sacrificing the scientific rigor gained from its underlying structure and principles.

The aim of this study was to develop and validate a method for estimating the Food Compass score for packaged foods and beverages using readily accessible nutrient and ingredient data available on the package in the United States and many other nations. This approach, referred to as Food Compass Score-10 (FCS-10), may serve as a practical solution to provide holistic and accurate assessment of healthfulness of branded foods and beverages for consumer communication and other applications.

METHODS

Food Compass scoring principles

The original Food Compass was developed based on an assessment of over 100 reported NPSs, a systematic review of national and international dietary guidelines, nutrient requirements for health claims, and an evaluation of food components linked to health outcomes.(5) In brief, the Food Compass incorporates 54 individual attributes across nine health-relevant domains. The 'nutrient ratios' domain includes measures of fat (unsaturated: saturated fat), carbohydrate (carbohydrate: fibre), and mineral quality (potassium: sodium). 'Vitamins' and 'minerals' domains focus on nutrients related to undernutrition and chronic disease. 'Food-based ingredients' includes foods with evidence for chronic disease impact. 'Additives' included factors with evidence for health harms (for example, nitrites in processed meats) and markers of adverse industrial processing (for example, artificial sweeteners, flavors or colors). 'Processing' incorporates the Nova classification, fermentation, and frying as emerging health-related processing factors. The remaining three domains incorporate specific lipids, fiber and protein, and phytochemicals. All attributes are assessed per 100 kcal (418.4 kj) of food product. Each domain's score is calculated as the average (or sum, for food-based ingredients) of its attribute scores, and then the domain scores are summed to calculate the Food Compass Score (FCS). The final FCS is scaled for interpretability to range from 1 (least healthful) to 100 (most healthful). In 2024, the Food Compass algorithm was updated based on updated evidence, availability of new data, and feedback from the scientific community.(7)

The domains, attributes, and scoring principles of the recently updated Food Compass algorithm have been published previously(7) and are outlined in **Supplementary Table 1**. In developing FCS-10, the underlying principles of the Food Compass were retained. Wherever data are available, known values are used to calculate the attribute scores for the Food Compass. We developed FCS-10 to estimate attribute scores where data are not available.

Test dataset

The study contains no human or animal data and no ethics approval was required. A test dataset was created of branded food and beverage products with a previously calculated full FCS and full Nutrition Facts and ingredient information to develop and evaluate FCS-10.

Branded products in the USDA Global Branded Food Products Database (GBFPD) were matched to the same product reported in the 2015-2020 Food and Nutrient Database for Dietary Studies (FNDDS), for which the full FCS had been computed. The GBFPD contains publicly available Nutrition Facts and ingredient list information provided by industry for over 400,000 branded and store-brand products, each identifiable by Global Trade Item Number (GTIN).(12) Details on the purpose and characteristics of the GBFPD are published in detail elsewhere.(13) The FNDDS contains comprehensive nutrient data (including values for 65 nutrients and food components) across all foods and beverages reported in What We Eat in America (WWEIA), the dietary intake component of the National Health and Nutrition Examination Survey (NHANES).(14) This facilitates the analysis of dietary intakes within these surveys. The 2015–2020 FNDDS corresponds to NHANES survey years 2015–16 through 2019–20. We utilized the GBFDB updated through October 2021 to match the FNDDS and minimize misclassification from reformulations that may have occurred thereafter.

Within the United States (US), a product's mandatory label lists serving size and nutrients per serving including energy, total fat, saturated fat, *trans* fat, cholesterol, sodium, total carbohydrate, dietary fiber, total sugars, added sugars, protein, iron, calcium, vitamin D and potassium; as well as all ingredients listed in descending order by weight.(15) Where matched GBFPD products were missing any mandatory attribute (generally <10%), we derived the missing value by direct review of product on the manufacturer's website or from the matched FNDDS product.

In identifying products for the test dataset, we started by looking for exact product matches for the 311 branded items reported within the FNDDS (see **Supplementary Figure 1**). Of the 311 branded items, 206 could be matched on both brand and product name (e.g., Hershey's Reese's Pieces) to a product within the GBFPD. To further expand the test dataset and include greater breadth of products, we next matched generic FNDDS products unlikely to differ substantially between brands to GBFPD products (e.g., a branded bagged spinach with no additions within the GBFPD was matched to the FNDDS product Spinach, raw). Products where some variation in composition was more likely were matched based on product name and available energy and nutrient data (e.g., Palak Paneer) to ensure a suitable match. Products were matched by two authors (E.W and Z.L) and checked by a third author (E.M.B). While certain food subcategories (e.g. cold cereals) had more matched products due to many FNDDS products in this category being branded, we ensured representation for each of 44 food and beverage subcategories created previously for use within FNDDS (**Supplementary Table 2**).(5) We excluded infant formula, baby foods, specialized dietary foods, alcohol, and products with <5 kcal/100g as these products are not intended to be scored by the Food Compass. Our final test dataset contained 538 branded foods and beverages (**Supplementary Figure 1**, **Supplementary Table 2**).

Food Compass 10 estimation

For 18 attributes with known quantities from nutrition facts panels or ingredients lists (e.g., fiber), we used these values to calculate the attribute score (**Supplementary Table 1**). This included additives, directly scored based on their presence in their ingredient lists. These known attributes contributed to scoring in 6 of the 9 Food Compass domains, including Nutrient Ratios, Vitamins, Minerals, Additives, Specific Lipids, and Fiber/Protein.

The ingredients list was used to score attributes in the Food Ingredients domain and to estimate missing nutrient values (e.g., vitamin B1) in other domains. For this purpose, we focused on the first five listed ingredients for any product, given the likely low caloric contribution for any additional ingredients. For composite ingredients having important sub ingredients (e.g., granola [rolled oats, sugar, honey]), the first three sub ingredients were separately extracted and utilized to define that ingredient. Each ingredient and sub ingredient was categorised into 1 of 168 ingredient categories adapted from 149 relevant WWEIA categories (**Supplementary Tables 3 & 4**).(16) Each of the 168 categories was then matched to a food-based attribute in the FCS (e.g. fruit, vegetable, whole grain, see **Supplementary Table 2**) or "other" for other ingredients (e.g., milk) (**Supplementary Table 4**). For scoring the Food Ingredient domain attributes, ingredients were scored as positive (e.g., fruits, vegetables, whole grains; score 10), negative (e.g., refined carbohydrates, processed or red meats; score -10), or non-scored (other ingredients; score 0). These domain calculations excluded trace ingredients, defined as ingredients categorised as "Additives, extracts, and emulsifiers" or "Spices including seasoning and salt", as well as any ingredients following these on the ingredients list (as these would be present in minimal amounts). The total Food Ingredient domain score for each product was calculated as the weighted sum of the first five ingredients' scores (including based on their sub ingredient scores, as applicable), with descending weights for each successive ingredient.

To calculate the weight to assign based on ingredient order, we let the first ingredient's weight be represented by x, and assigned all subsequent ingredients 2/3 of the weight of the previous ingredient (up to a maximum of five ingredients). We then let the sum of all ingredients weights equal to 100, and solved for X, as shown in equation 1:

$$\sum_{n=1}^{5} x(\frac{2}{3})^{n-1} = 100$$

Equation 1. Mathematical approach used to assign weights to ingredient scores, where n is the number of ingredients and x is the weight applied to the first ingredient's attribute score.

Therefore, for a food with four non-trace ingredients, the Food Ingredient domain score of the 1st, 2nd, 3rd, and 4th ingredient would contribute approximately 42%, 28%, 18% and 12%, respectively to the foods final Food Ingredient score.

The ingredients list was also used to estimate missing attributes in the Vitamin, Mineral, Specific Lipids, and Phytochemical domains. Based on complete FCS data from 9,767 unique foods and beverages in FNDDS, a mean (reference) FCS score for each attribute (e.g., vitamin E) was calculated for each of the 168 food and beverage ingredient categories (**Supplementary Table 3**). Each branded product's missing attribute score was then estimated based on the weighted, summed score of the reference attribute score corresponding to each of its first five listed ingredients, using the same approach (**Equation 1**), with the same principles applied for parenthetical ingredients and trace ingredients. A complete example of the ingredient-based approach is presented in **Supplementary Methods 1**.

Food products were classified into Nova categories using the ingredient lists and specific keywords. We followed established methods for categorizing foods according to Nova classification.(17) Items were designated as ultra-processed (Nova 4) if they contained key additives such as emulsifiers, artificial sweeteners, partially hydrogenated oils, high fructose corn syrup, monosodium glutamate, or processed ingredients (e.g. xylitol). Products not meeting the criteria for ultra-processed were further categorized based on their ingredients. Products containing multiple key ingredients like oil, fats, sugar, or salt, including items like cheeses, canned vegetables or seafood, salted nuts, and fruit preserved in syrup, were classified as processed foods (Nova 3). These items often include added salt or sugar for preservation. Products with a single key ingredient such as oil, fats, sugar, or salt, and not classified as processed or ultra-processed, were categorized as processed culinary ingredients (Nova 2). These items include cooking staples like oils, sugar, vinegar, honey, and butter, which typically undergo minimal processing like pressing or milling but contain few additives. Foods classified as unprocessed or minimally processed (Nova 1) were those with minimal ingredients, lacking added salt, sugar, or fats. These products were primarily whole foods with minimal processing and not already flagged as Nova 2 or Nova 3. Final classifications were manually reviewed and adjusted for consistency and accuracy based on a comprehensive keyword list.

Products were scored as containing nitrites (binary) if they were categorised within the "cured meat" food category or contained key terms (e.g., bacon, cured, nitrite, salami, sausage, smoked; see **Supplementary Methods 2** for full details) in their product name or first 5 ingredients. Products were scored as fermented according to their food group, including yogurt, cheese (except plant-based cheeses)(18), and other products with key terms in their description or first five ingredients (e.g. culture, dosa, fermented, injera, kefir, kimchi, kombucha, miso, natto, probiotic, see **Supplementary Methods 2** for full list). Products were scored as fried if they contained key terms in their description (e.g. battered, breaded, deep-fried, egg roll, fried, fritter, potato chips, sauteed, spring rolls, tempura, see **Supplementary Methods 2** for full list) and also contained oil or fried in their first three ingredients.

Once all attribute and domain scores were estimated, the FCS-10 was calculated using the standard Food Compass algorithm, whereby each domain's score is calculated as the average of its attribute scores, and then the domain scores were summed to calculate the

FCS (as described in **Supplementary Table 1** and detailed previously)(7). Acknowledging the lower precision of this approach, the final FCS-10 was scaled from 1 (least healthful) to 10 (most healthful), as opposed to the more granular range of 1 to 100 used in the standard FCS.

Validation

After rounding the original FCS (published previously)(7) to range from 1 to 10, we compared the FCS-10 and FCS for all matched products within the test database. Root mean squared error (RMSE) and Spearman's rank correlation (r) were calculated to assess the performance of FCS-10 overall and by food category. Classification accuracy was calculated as the proportion of correctly predicted FCS scores, correctly predicted within ±1 unit, and correctly predicted by recommendation category cut-offs. We used similar cut-offs to those previously proposed for the FCS to categorise products as foods to be encouraged (FCS-10 ≥7), foods to be consumed in moderation (FCS-10 = 4-6) and foods to be minimized (FCS-10 ≤3).(7) Scatter plots were inspected to visually compare FCS-10 to FCSX (where y=x depicts perfect scoring).

We performed a diagnostic accuracy analysis to evaluate the performance of the FCS-10 compared to the FCS. Sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were calculated by comparing the FCS-10 recommendation categorisations with the FCS recommendation categorisations (≥7, 4-6, ≤3). Sensitivity was defined as the proportion of true positives correctly identified by FCS-10, while specificity was defined as the proportion of true negatives correctly identified. PPV was calculated as the proportion of true positives among all positive predictions, and NPV was the proportion of true negatives among all negative predictions. Overall sensitivity, specificity, PPV and NPV for the FCS-10 were calculated using the micro-average method. All analyses were conducted using Stata 18 (StataCorp, College Station, TX) and R Version 4.4.1 (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

A test dataset was created of branded food and beverage products with a previously calculated and published full FCS (7) and full Nutrition Facts and ingredient information to develop and evaluate FCS-10. The FCS and FCS-10 for all 538 products in the dataset are shown in **Supplementary Table 2.** The FCS-10 approach predicted 49% (n=261) of product scores exactly, 89% with at most 1 unit difference (n=481), and 100% (n=538) with at most 2 units difference. By food category, the percentage of products with a deviation of 2 units from the FCS ranged from 0% for seafood to 22% for sauces and condiments. The performance of FCS-10 on the test dataset is displayed in a scatter plot (**Figure 1**); systematic over- or underestimation was not evident.

The FCS-10 predicted FCS with good accuracy overall (0.90 root mean squared error (RMSE)) and across most food categories (all <1.0 RMSE except for sauces and condiments) (**Table 1**). The FCS-10 also accurately ranked products from highest to lowest FCS (Spearman's r=0.93). FCS-10 appeared most apt at estimating the standard FCS for fruit (0.61 RMSE, r=0.90), fats and oils (0.68 RMSE, r=0.94), meat, poultry, and eggs (MPE; 0.78 RMSE, r=0.89), and vegetables (0.79 RMSE, r=0.93). For these food categories, the FCS-10 accurately predicted the standard FCS within ±1 units for 97%, 96%, 95% and 93% of products, respectively. The FCS-10 was still reasonable but relatively less likely to estimate the standard FCS for sauces and condiments (1.01 RMSE, r=0.86) and legumes and nuts (0.99 RMSE, r=0.75); for these categories it accurately predicted the FCS within ±1 units for 78% and 84% of products, respectively. For seafood, the FCS-10 only exactly estimated the FCS of products 44% of the time, but no estimated scores deviated by more than 1 unit from the standard FCS (RMSE 0.60).

After grouping by recommended FCS thresholds (≤3 for foods to be minimized, 4-6 for foods to be consumed in moderation, ≥7 for foods to be encouraged), FCS-10 correctly classified 87% of products. The group classification accuracy ranged from 100% for seafood products to 81% for grain products. Other food categories with greater than 90% group classification accuracy included fruit (97%), fats and oils (96%), and savory snacks and sweet desserts (92%) (**Table 1**).

The FCS-10 was slightly more apt at correctly classifying products as foods to limit (sensitivity 87%, specificity 96%, positive predictive value (PPV) 91%, negative predictive value (NPV) 94%) and foods to encourage (sensitivity 89%, specificity 96%, PPV 89%, NPV 96%) compared to foods to eat in moderation (sensitivity 85%, specificity 88%, PPV 81%, NPV 90%) (**Table 2**). The overall sensitivity, specificity, PPV and NPV of FCS-10 were 87%, 93%, 87% and 93%, respectively.

DISCUSSION

In this research, we describe and validate a novel method, FCS-10, for estimating the Food Compass using commonly available (and mandatory in numerous nations) nutrient and ingredient list information. Among 538 diverse foods and beverages, FCS-10 produced scores within 1 unit of the standard FCS for ~90% of products. FCS-10 also performed well in identifying products to limit (≤3), moderate (4-6), or encourage (≥7), with overall sensitivity and specificity of 87% and 93%, respectively. This approach demonstrates reasonable validity of FCS-10 as a pragmatic solution to evaluate the healthfulness of food and beverage products in-store, online, or in any food dataset that includes only typically available nutrient and ingredient data. Given all data needed are publicly available, food retailers and manufacturers could also use the FCS-10 to evaluate the healthiness of their own portfolios or product range.

Existing NPSs used globally, such as the Health Star Rating in Australian and New Zealand and the Nutri-Score in Europe, are based on mostly available nutrient data and perform reasonably well.(19) However, they also have some limitations, including requiring variable scoring across different food and beverage categories, omitting key characteristics of importance to health like processing, penalizing of healthy fats by penalizing energy density, insufficient discrimination between whole grain and refined cereal products, and using weight-based scoring which causes inconsistencies due to varying water content. The standard Food Compass overcomes each of these limitations but requires data on several attributes that are not commonly available. FCS-10 offers a solution by enabling the estimation of FCS using commonly available information, optimizing its practical utility while maintaining the strengths of Food Compass foundational structure and principles.

The FCS-10 was most successful at estimating accurate scores for fruits, vegetables, meats, poultry, eggs, and fats and oils. These foods generally contain fewer ingredients and are less likely to have dramatic variation in their nutrient composition. Yet even for more complex products within these categories such as stuffed olives, creamed spinach, or sweet potato chips, the FCS-10 was accurate. Despite containing more ingredients, these products appear to maintain a more consistent nutrient profile across different brands and recipes, likely owing to having one main ingredient (e.g. olives, spinach, sweet potato, respectively).

FCS-10 performed least well for sauces and condiments, where 78% of products had a score within 1 unit of the standard FCS. This may be explained by the large variation in ingredients and nutrients within this category. Ma et al. 2021 used machine learning approaches to estimate mandatory labelled nutrients for products in the GBFDB and found that their model performed comparatively less well for condiments.(20) The variability in nutrients can be large. For example, the energy density of buffalo sauce products in the GBFDB ranged from 48kcal/100g to 233 kcal/100g, despite having similar main ingredients (cayenne peppers, water, vinegar, oil, salt). Variations in proportions of these ingredients can impact the products' nutritional profiles considerably. In the future, if ingredient proportion information were available for even the top ingredients, more accurate weighting would undoubtedly improve the precision of our method for sauces and condiments.

In developing FCS-10, we prioritized distinguishing between whole- and refined-grain ingredients, as these attributes are positively and negatively scored in Food Compass, respectively. We disaggregated the existing WWEIA grain ingredient categories accordingly. In contrast, some WWEIA ingredient categories, such as "nuts and seeds" and "fish," remained relatively broad. This may partly explain why FCS-10 was less precise in predicting scores for legumes and nuts and seafood, where foods are largely composed of these ingredients. Disaggregating these ingredient categories could potentially improve accuracy. However, unlike cereal products, where nutritional differences are more pronounced, the standard FCS within these categories are relatively concentrated. All products within the legumes and nuts and seafood categories in the test dataset scored between 7 and 10, with the majority receiving scores of 9 or 10. Therefore, while exact classification accuracy appeared lower for these categories than for others, accuracy within one score was very high, as was accuracy within the broader group classification (i.e., foods to encourage).

While machine learning had similar challenges in scoring certain categories like sauces, it may become useful for estimating certain attributes, such as Nova processing.(21) Other studies have applied machine learning techniques to estimate nutrient content and food categorization from other available nutrient, ingredient and similar product data with fairly high accuracy.(20, 22-25) Automated electronic coding of ingredient lists can be challenging due to variations in names for similar ingredients (e.g., high fructose corn syrup and glucose-fructose syrup; non-fat milk and skim milk; bell pepper, capsicum and sweet pepper), spelling differences and errors, and even omitted punctuation (such as commas between ingredients). Integrating machine learning approaches into FCS-10 could improve the speed and accessibility of the method. Use of artificial intelligence large language models could also facilitate the recognition and interpretation of ingredients lists.

Our study has several strengths. We developed and described a novel approach to estimate FCS for packaged food products based on readily available nutrient and ingredient list information. We confirmed validity using 538 matched products with known FCS. The accuracy of our approach was assessed overall and at the food category level using multiple performance metrics, providing a comprehensive evaluation of its performance across a variety of foods, including those with more complex compositions. The results demonstrated similarly high performance across most food categories. The intra-category discrimination of the FCS-10 was evident when assessing the range of FCS-10 scores for foods within categories with highly variable nutritional quality. For example, cold cereals scores included 7 for a high-fibre, low-sugar, non-ultra-processed, fortified cereal and 1 for a high-sugar, high-refined starch, ultra-processed cereal. Our method is publicly available, straightforward to apply with modern technology and coding, and does not require specialized statistical expertise, increasing applicability by a wide range of stakeholders.

As a limitation, we did not have large numbers of products in every food category, limiting precision of our validity analysis in subcategories. The task of categorizing ingredients for large food datasets is labor-intensive. Overcoming this limitation, such as through machine-learning approaches described earlier, would allow our method to be more automated. The reference dataset for the present investigation was derived from US data, which may be reasonably representative of multinational packaged foods worldwide but less so for locally produced products in other countries, or multinational packaged food with differing compositions globally. Yet, given the elemental nature of most of the 168 ingredient categories, the increasingly global nature of food supplies, and the typical convention of labelled ingredients listed in descending order, FCS-10 may still offer reasonable approximations for packaged food products in most contexts. Although not essential prior to its application, testing FCS-10 in other regions with differing available food products, and validating FCS-10 against health outcomes would further strengthen its validity.

In summary, this investigation presents the methods and validation for a new approach to estimate FCS for foods and beverages using commonly available nutrient and ingredient information when the data required to calculate the original Food Compass are not available. This practical approach is ready for immediate use and can be applied by researchers, policymakers, retailers, and manufacturers to assess the healthfulness of packaged foods and beverages for a broad range of applications.

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Data availability

The attribute and domain scoring algorithm used to generate the Food Compass is available in Supplementary Table 1. The reference attribute scores, ingredient categorization, and ingredient weighting approach needed to generate FCS-10 for the test dataset are available in Supplementary Table 3, Supplementary Table 4 and Methods/ Supplementary Methods 1, respectively. All data used in this analysis is publicly available from the following USDA sources: (1) 2015-2020 Food and Nutrient Database for Dietary Studies (FNDDS) [https://www.ars.usda.gov/northeast-area/beltsville-md-bhnrc/beltsville-human-nutrition-research-center/food-surveys-research-group/docs/fndds-download-databases/], (2) Branded products in the USDA Global Branded Food Products Database (GBFPD) [https://fdc.nal.usda.gov/download-datasets.html]. The generated Food Compass and FCS-10 for each of the 538 items in the test dataset are available in Supplementary Table 2.

Competing interests

Dariush Mozaffarian reports financial support was provided by National Institutes of Health. Eden M Barrett reports financial support was provided by National Health and Medical Research Council. Jeffrey B Blumberg reports a relationship with Guiding Stars Licensing Company that includes: consulting or advisory. Dariush Mozaffarian reports a relationship with Bergen Therapeutics that includes: board membership. Dariush Mozaffarian reports a relationship with Brightseed Inc that includes: board membership. Dariush Mozaffarian reports a relationship with Calibrate that includes: board membership and equity or stocks. Dariush Mozaffarian reports a relationship with Elysium Healthcare Limited that includes: board membership. Dariush Mozaffarian reports a relationship with Filtricine Inc that includes: board membership. Dariush Mozaffarian reports a relationship with HumanCo that includes: board membership and equity or stocks. Dariush Mozaffarian reports a relationship with Instacart Health that includes: board membership. Dariush Mozaffarian reports a relationship with January Inc. that includes: board membership. Dariush Mozaffarian reports a relationship with Season Health that includes: board membership. Dariush Mozaffarian reports a relationship with Validation Institute that includes: board membership. Dariush Mozaffarian reports a relationship with WndrHlth that includes: board membership. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 1. Performance of the Food Compass Score-10 (FCS-10) approach on the test dataset, shown overall and within 12 major food groups.

Food category	n	FCS-10 (mean±SD)	FCS (mean±SD)[1]	Root mean squared error[2]	Spearman correlation[3]	Classification accuracy (exact)[4]	Classification accuracy (±1 unit)	Classification accuracy (recommendation)
OVERALL	538	5.0 ± 2.7	5.0 ± 2.7	0.90	0.93	0.49	0.89	0.87
Grains	178	4.6 ± 2.0	4.5 ± 1.9	0.97	0.85	0.42	0.87	0.81
Beverages	26	3.8 ± 2.5	3.8 ± 2.5	0.97	0.93	0.58	0.85	0.88
Fruit	35	8.7 ± 1.8	8.5 ± 1.9	0.61	0.90	0.69	0.97	0.97
Vegetables	41	8.1 ± 2.1	8.4 ± 1.9	0.79	0.93	0.59	0.93	0.88
Legumes and nuts	19	8.4 ± 1.6	8.8 ± 1.0	0.99	0.75	0.37	0.84	0.84
Meat, poultry and eggs	40	4.1 ± 1.8	4.1 ± 1.8	0.78	0.89	0.55	0.95	0.85
Seafood	9	8.7 ± 0.7	8.3 ± 0.9	0.60	0.48	0.44	1.00	1.00
Dairy	25	5.3 ± 1.7	5.1 ± 1.6	0.91	0.76	0.44	0.92	0.88
Fats and oils	23	5.1 ± 2.5	5.2 ± 2.5	0.68	0.94	0.70	0.96	0.96
Mixed dishes	48	4.5 ± 2.1	4.4 ± 2.3	0.95	0.86	0.38	0.85	0.83
Sauces and condiments	23	4.3 ± 2.5	4.3 ± 3.0	1.01	0.86	0.26	0.78	0.87
Savory snacks and sweet desserts	71	2.4 ± 1.7	2.4 ± 1.5	0.85	0.81	0.55	0.92	0.92

1. Original FCS was rescaled to run from 1 (lowest) to 10 (highest) to compare to FCS-10.

2. Root mean squared error (RMSE) is the standard deviation of the residuals. It quantifies the prediction accuracy of FCS-10, with lower values indicating better fit and smaller differences between predicted FCS-10 values and true FCS values. Compared to a simple mean absolute error, which treats all deviations equally, RMSE gives more weight (and is therefore more sensitive) to larger deviations.

3. Spearman rank correlation measures how accurately the FCS-10 approach ranks products from highest to lowest FCS, with a value of 1 indicating perfect ranking.

4. Classification accuracy measures the proportion of correctly predicted FCS scores (exact), correctly predicted within ±1 unit, and correctly predicted FCS recommendation category (≤3 for foods to be minimized, 4-6 for foods to be consumed in moderation, ≥7 for foods to be encouraged).

Table 2. Sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) of the Food Compass Score-10 (FCS-10) as compared to the standard Food Compass Score (FCS) by categories to encourage, moderate, and minimize[1]

Panel a. Classification of products as foods to be encouraged

	Standard FCS			
FCS-10	≥7	<7		
≥7	133	16	89%	PPV
<7	17	372	96%	NPV
	89%	96%		
	Sensitivity	Specificity		

Panel b. Classification of products as foods to be consumed in moderation

	Standard FCS			
FCS-10	4-6	≤3 or >6		
4-6	175	40	81%	PPV
≤3 or >6	32	291	90%	NPV
	85%	88%		
	Sensitivity	Specificity		

Panel c. Classification of products as foods to be minimized

	Standard FCS			
FCS-10	≤3	>3		
≤3	158	16	91%	PPV
>3	23	341	94%	NPV
	87%	96%		
	Sensitivity	Specificity		

1. FCS threshold groups defined as foods to be encouraged (≥7; panel A), foods to be consumed in moderation (4-6; panel B) and foods to be minimized (≤3; panel C).
Sensitivity measures the proportion of actual positives that were classified correctly and is calculated using the formula: true positives/(true positives + false negatives).
Specificity measures the proportion of actual negatives that were classified correctly and is calculated using the formula: true negatives/(true negatives + false positives).
Positive predictive value (PPV) measures the proportion of positive classifications that were correct and is calculated using the formula: true positives/(true positives + false positives).
Negative predictive value (NPV) measures the proportion of negative classifications that were correct and is calculated using the formula: true negatives/(true negatives + false negatives).

Figure Legend

Figure 1. Scatter plot of Food Compass Score-10 (FCS-10) predictions compared to true Food Compass Score (FCS) on the test dataset (n=538). Spherical random noise has been added to illustrate density of data. Original FCS was rescaled to run from 1 (lowest) to 10 (highest) to compare to FCS-10.